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ORIGINAL ARTICLE

An experimental investigation of soft price caps in uniform price auction markets for wholesale electricity

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Abstract This paper presents the results from an auction experiment using industry professionals and student participants who compete in a simulated wholesale market for electricity. Motivated by the intervention by FERC in response to the "meltdown" of the California spot market, we investigate the effect of including a "soft" price cap in a uniform price auction as a means of mitigating high prices. When prices are driven above the soft cap, offer curves become flat, in contrast to the hockey stick shape observed in a typical uniform price auction for electricity. This flat offer curve leads to market prices that are relatively insensitive to both generation costs and demand reductions.

Keywords Soft cap auction · Uniform price auction · Electricity markets · Experimental auctions

JEL Classifications L94 · D44 · C92

1 Introduction

Price caps continue to be widely used as a price mitigation measure in deregulated wholesale markets for electric power. Perhaps the most debated and publicized price

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cap interventions occurred in the midst of the California electricity crisis, spanning May 2000 to June 2001. The high prices in summer 2000 motivated the California Independent System Operator (CAISO) to lower the price cap from $750/MWh to $250/MWh. The Federal Energy Regulatory Commission (FERC) later declared that the spot market was dysfunctional and ordered the price cap be replaced by a new type of "soft" price cap, implemented in December 2000 at $250/MWh, and then revised in January 2001 to $150/MWh. The soft price cap effectively converted the existing uniform price auction to one that was a hybrid of a uniform price and a discriminatory auction. In particular, in what we hereafter refer to as a "soft cap auction", all offers to sell generating capacity below the soft cap are submitted in a uniform price auction, and paid a clearing price equal to the last (highest) accepted offer from those submitted below the soft cap.[1] Any capacity offered above the soft-cap needed to meet load (i.e., demand) is paid the actual offer in a discriminatory price auction. Offers above the soft cap are subject to cost-justification.[2] Nevertheless, high prices persisted during the winter and spring of 2001. In June 2001, FERC implemented a western US-wide mitigation procedure, which included a $91.87/MWh soft price cap. The soft cap has remained in place, although has been raised to $500/MWh and is now commonly referred to as a "damage control bid cap", and since October 2002 has been used in tandem with an automated mitigation procedure (AMP).

This paper presents results from an experiment that investigates the performance of uniform price and soft cap auctions, with and without price-responsive load. Both industry professionals and students are used as participants who represent generators in a "smart" market, PowerWeb, which replicates the physical constraints of meeting loads on an electrical grid. The first objective of this paper is to explore whether a soft cap is capable of mitigating high prices, in a situation when prices are initially driven above the soft cap. The second objective is to investigate whether the inclusion of a soft cap affects the effectiveness of a price-responsive load initiative.

2 Related literature

Most theoretical and experimental work on private-value auctions focuses on the case where each seller (buyer) has single unit supply (demand) (see, for example, Cox et al. 1984, 1985; Harris and Reviv 1981). With single unit supply, it is well-known that the uniform price auction is efficient, and there is a weakly dominant strategy for sellers to submit an offer equal to cost. With risk-neutral agents, the discriminatory auction is also efficient and expected revenue is equivalent for discriminatory and uniform price auctions (Harris and Reviv 1981). These properties do not extend to a situation with multiunit supply, which characterizes wholesale markets for electric power.

[1] Consistent with the auction literature, we use "offers" to describe seller auction price submissions. We note, however, that these offers are referred to as "bids" in many naturally occurring wholesale markets for electric power.

[2] However, FERC has been criticized for undertaking a very cursory review of cost justifications (Wolak 2003) and empirical evidence suggests that firms were able to justify costs by manipulating natural gas input prices (Jacobsen and Shaikh 2004; Wolak 2003). We do not include any cost-justification procedure in our experimental design.

Krishna (2002) offers a clear presentation of the theory of multiunit uniform price and discriminatory auctions.[3] In particular, when there are multiple sellers who each submits price offers for multiple units, pure strategy equilibria are known to exist for both auction formats, but no closed form expressions for equilibrium strategies are available. Further, both auctions are inefficient. Seller behavior in a uniform price auction is characterized by an offer equal to cost for the lowest-cost unit, with offers greater than cost for all other units. A discriminatory auction seller has a positive probability of submitting a flat offer schedule, in the sense that offers for all units are equal, and some probability of submitting one that is upward sloping. Fabra et al. (2006) provide additional insight on these two auctions in the particular context of electricity auctions. Of particular relevance for our study, their theoretical analysis suggests that, with uncertain demand, expected demand changes will have a greater effect on prices in a uniform auction.

A handful of papers using experimental methods, in particular those related to wholesale electricity markets, find auction behavior that is largely in agreement with theory. In particular, Bernard et al. (2005) and Mount et al. (2001), in uniform price auctions with six sellers and stochastic but perfectly inelastic demand, find that sellers tend to submit offers close to cost for lower cost units and speculate with or withhold marginal, high cost units. When low-cost units represent a significant share of offered capacity, the resulting offer curve takes the shape of a hockey stick. Reports from FERC, CAISO, PJM and other wholesale markets commonly refer to offer curves having this shape (see, for example, Marcus and Ruszovan 2002). An example of a typical offer curve in the PJM Interconnection is shown in Fig. 1. Offer curves in the discriminatory auction experiments of Mount et al. (2001) and Rassenti et al. (2003a) are generally upward sloping, but rather flat, with offers considerably higher than uniform price offers for lower-cost units.

The results of earlier tests of electricity markets conducted with PowerWeb show that multiunit uniform price and discriminatory auctions can produce average prices 50% above competitive levels (Mount et al. 2001). However, the prices for the uniform price auction are more volatile with many price spikes. This volatility is consistent with price patterns in naturally occurring markets. As an illustration, the daily spot prices and maximum load for the PJM Interconnection are shown in Fig. 2. With the hockey stick shaped offer curve, small fluctuations in load can result in large price changes as the last accepted offer sets the price for everyone. With the rather flat offer curves typical of discriminatory price auctions, fluctuations in load decrease the sensitivity of (average) market price to fluctuations in load. This flat shape is likely to undermine the effectiveness of demand conservation as a way of mitigating high prices.

Experimental investigations of price mitigation measures have largely been in the form of introducing price-responsiveness in load.[4] Mount et al. (2001), through

[3] Krishna's discussion is in terms of buyer behavior in the case of fixed supply, so we draw analogies for the analytically equivalent case of seller behavior with fixed demand.

[4] Notable exceptions include Kiesling and Wilson 2007, who explore an AMP with features similar to that currently in place in the California spot market, Chapman et al. (2004), who investigate the effects of market size and communication on prices, and Brandts et al. (2008), who study the effects of allowing for forward contracts between firms and traders.

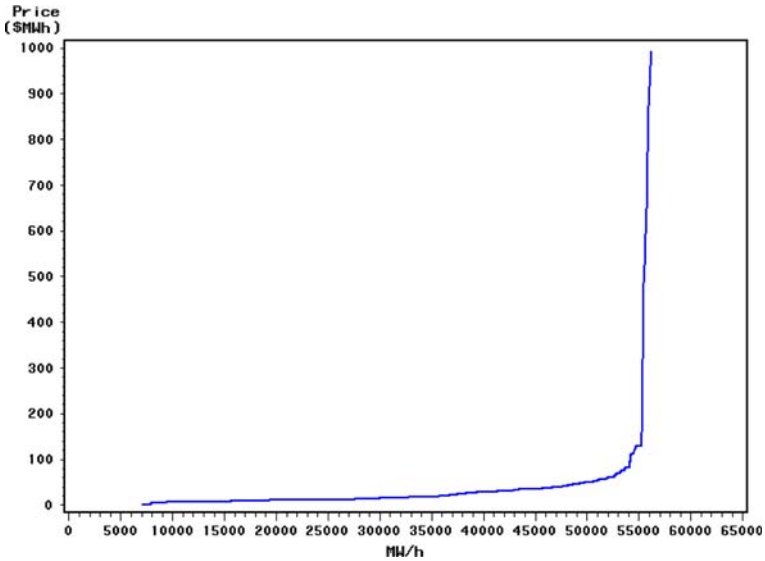

Fig. 1 A typical aggregate offer (supply) curve in PJM

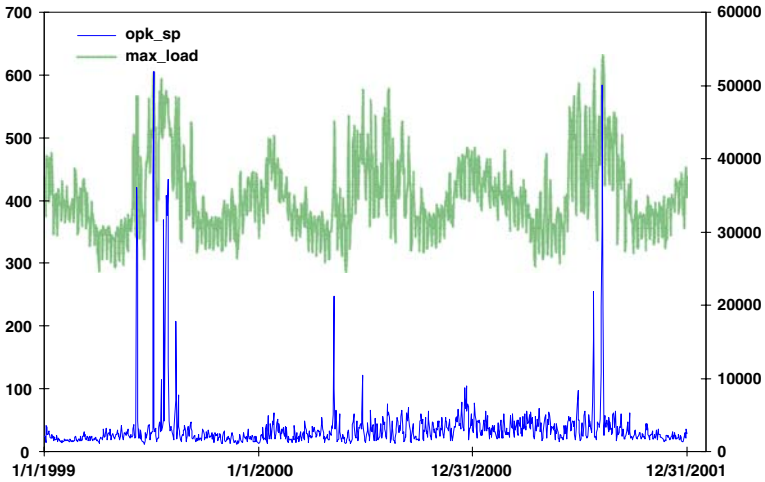

Fig. 2 Daily maximum load and on-peak price in PJM

contracts for interruptible load, find that prices are reduced by 13% for a uniform price auction. With a related but different experimental design, Rassenti et al. (2003a) find that discriminatory price auctions produce much higher prices than uniform price auctions in treatments with price-responsive load. They show that (1) price volatility is reduced in a discriminatory auction, and (2) prices are considerably above competitive levels with both types of auctions.[5] In a set of related experiments, Rassenti et al.

[5] Abbink et al. (2003), however, find that price volatility is similar across uniform price and discriminatory auctions when information about demand is asymmetric.

(2003b) find that the introduction of demand-side bidding—where human participants submit bids to buy power, and thus make the market less dominated by sellers—effectively brings prices down to competitive levels. Chao and Plott (2000); Denton et al. (2001) and Nicholaison et al. (2001) describe and test two-sided auctions for electricity that produce competitive outcomes.

Hence, there is a growing body of literature that suggests wholesale markets for electricity would be more competitive if demand responded to price. In contrast, there is relatively little research on the relative performance of using a soft cap auction instead of a pure uniform price auction in a wholesale market for electricity. Jacobsen and Shaikh (2004) develop a theoretical model of a monopolistic firm bidding in a soft cap auction for electricity. They find that under reasonable conditions, the introduction of a soft cap provides strict incentives for firms to bid above the soft cap as well as increase capacity withholding. They find theoretically that firms have incentives to cost-justify offers above the soft cap through the distortion of electricity input prices, which is required by the CAISO, and provide empirical evidence of this occurrence using data from the natural gas spot market and the southern California NO_x emissions permit market.

3 Experimental design

3.1 Experimental framework

This study utilizes PowerWeb, an interactive, distributed, Internet-based simulation environment for testing various auction markets using human decision makers.[6] PowerWeb assumes the presence of a central agent acting as an independent system operator (ISO) to assure the reliable operation of the physical power system. The PowerWeb environment is designed to run unit commitment and optimal power flow routines to provide generation schedules that minimize the cost of meeting load subject to the physical constraints of an AC network. However, for the experimental auctions discussed in this paper, network constraints are not binding.

Figure 3 summarizes the design of our experimental market. Each participant represents one of six firms in an electricity market. Each firm owns five generating units and has 100 MW of total capacity. The baseload generator is 50 MW and operates at a cost of $20 per MWh. The second generator is 20 MW with a marginal cost of $40/MWh. The last three generators are 10 MW each with marginal costs of $48/MWh, $50/MWh and $52/MWh.

On the demand side, in each trading period, the generator incurs a fixed interest charge of $1200 (to make earnings roughly equal to profits in excess of competitive levels) and is given a forecast of the system load. The forecasted load in each period is randomly generated using a uniform distribution within a band of 430–550 MW. On average, demand is 490 MW or 82% of available capacity. Actual load in each period is equal to forecasted load plus a small stochastic error within the range ±20 MW (about ±4%). For most periods, some of the marginal (expensive) generators are needed to

[6] The PowerWeb homepage is currently at www.pserc.cornell.edu/powerweb/.

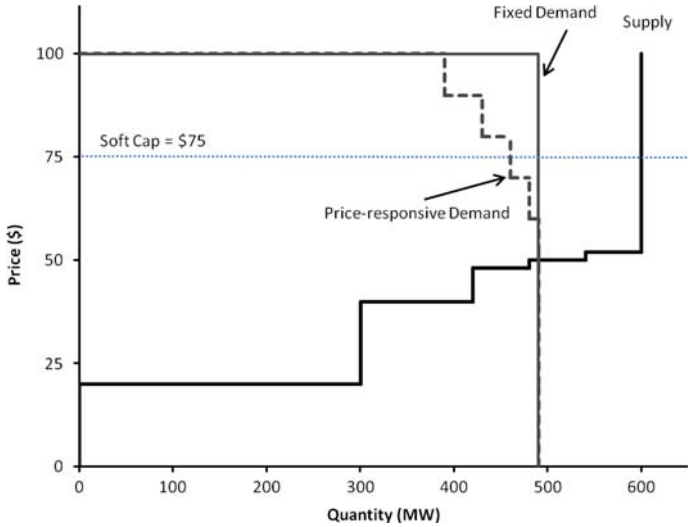

Fig. 3 Experimental market design with baseline costs

meet load. This design is thus intended to characterize supply behavior during periods of shoulder and peak demand.

In treatments without price-responsive load, the demand curve is perfectly inelastic, as depicted by the "Fixed Demand" array in Fig. 3 for a case where actual load is 490 MW. In treatments with price-responsive load, there are contracts for interruptible load. This type of contract was used in California during the early stages of soft cap auction use. Contracts are predetermined, do not vary across periods, and participants do not have details of the contracts. Contracts exist at $60, $70, $80, and $90 for 10, 20, 30, and 40 MW of interruptible load, respectively. The resulting "Price-responsive Demand" array is illustrated in Fig. 3 for a load of 490 MW.

As an example, suppose there are not enough offers below $60/MWh to meet the load. The ISO exercises the contract and the effective load is now up to 10 MW less, depending on whether the load can be met using the contract. If the contract is fully exercised and offers below $70 are not sufficient to meet the modified load, the second contract is partially or fully exercised (up to an additional 20 MW). Additional contracts are exercised as needed. At most load can be reduced by 100 MW, equivalent to the total capacity of one firm.

In each dispatch period, the firm submits offers to sell capacity into an auction run by an ISO. A separate offer is made for the capacity of each generating unit submitted, and the maximum offer allowed (i.e., a hard price cap) is $100/MWh.[7] A stand-by cost of $5/MWh is incurred for all offered capacity, regardless of whether an offer is accepted. This stand-by cost is included to represent the opportunity cost of being available for a time period, foregoing sales in other markets or delaying maintenance

[7] Our price cap is relatively low in order to reduce the variability in participant earnings, and to keep the maximum possible payment to participants reasonable.

activities. The firm can choose to withhold a generator to avoid the stand-by cost and this makes participants consider the issue of withholding capacity.

The ISO selects the least expensive combination of offers to meet the system load. If insufficient capacity is submitted to meet load, the ISO randomly recalls generating units withheld from the auction. The firm is charged $10/MWh for each recalled generator and receives a price equal to the highest offer in the original auction. Once the results have been reviewed by the firms, the process is repeated until the end of the session when a predetermined number of periods have been completed.

3.2 Experiments and hypotheses

With uniform price and soft cap auctions, with and without price-responsive load, there are four distinct treatments. All treatments have in common the features described above. Both the uniform price and soft cap auction select the lowest cost combination of offers to meet load. For the uniform price auction, the market clearing price is equal to the last accepted offer. The soft cap in the soft cap auction is set to $75. Offers less than the soft cap are submitted into a uniform price auction, where the last accepted offer below the soft cap sets the market price for successful offers. Offers greater than the soft cap are submitted into a discriminatory auction, and any accepted offer is paid a price equal to the actual offer amount.

The costs for the four smallest generators (50% of capacity) varied during each session with student participants to reflect changes in input costs such as natural gas prices. The costs of the three smallest blocks varied for the industry professionals. Approximately 50% of electric power generation comes from natural gas in California. Prices of natural gas on the spot market at Henry Hub began rising in May, 2000 and peaked in December, 2000. In addition, there was a dramatic increase in the reported prices of natural gas delivered in California in December 2000 and January 2001. This was exactly when the soft-cap auction was implemented. In January, 2001, the spot price for natural gas in California may have been over ten times greater than it was at the same time in the previous winter.[8]

Our firms experienced a pattern of both high and low generation costs for the more expensive generators in each session. We implemented this to observe whether electricity prices declined with marginal costs. The pattern of generation cost was similar across experiments.[9] At or near the beginning of each experiment, costs for the relevant generating blocks were increased by $30/MWh. High costs persisted for 10–15 periods with students (5–10 with professionals), and then decreased by $5/MWh each dispatch period for five periods. After this, the baseline costs stayed in effect for the remainder of the experiment. An important implication of this cost increase is that it

[8] A staff report from FERC (2002) shows how these delivery prices may have been manipulated. The report recommended that the reported prices of natural gas should not be used for paying refunds in California. Also, as described above, Jacobsen and Shaikh (2004) provide empirical evidence of manipulation in the natural gas spot market.

[9] The pattern varied as, in the treatments with professionals as described below, these professionals competed in the same group. It was our intention to not make the pattern of costs predictable, so that participants would pay close attention to their costs before submitting prices offers in a given period.

forces all these generators to have costs above the $75 soft cap. Thus, as at least some capacity is needed from these generators, the competitive market price in turn is above the soft cap.

Experience from experimental and naturally occurring markets suggests that uniform price offer curves have a hockey stick shape and discriminatory offer curves are relatively flat. Based on the theoretical analysis of Jacobsen and Shaikh (2004) for the soft cap auction, and due to the fact that high initial generation costs should drive offers into the discriminatory auction segment of the soft auction, we expect soft cap offer curves similar to pure discriminatory auction offer curves. With these expected shapes of offer curves, we have two hypotheses: (Hypothesis 1) price-responsive load will lead to a larger price decrease in the uniform price auction; (Hypothesis 2) a cost decrease (increase) will lead to a larger price decrease (increase) in the uniform price auction. As discussed by Abbink et al. (2003), there is mixed evidence whether discriminatory or uniform price auctions generate high prices. Accordingly, we have no expectation about the relationship between uniform price auction and soft cap auction prices.

3.3 Participants

We conducted four sessions—one corresponding with each treatment—using undergraduate business and economics students at SUNY-Binghamton. Sessions were conducted in a designated computer laboratory. There were 18 participants in a session, who competed in three distinct markets with six sellers. Each participant was paid in direct proportion to her generator's earnings. There were 50 trading periods in each session. On average students were paid $25 for a session that lasted between 60 and 90 min.

Using 18 individuals from the electric power industry, we also conducted one session wherein participants played 25 periods of the soft cap auction without price-responsive load followed by 25 periods of the soft cap auction with price-responsive load. Participants were recruited via email and worked as traders, consultants, analysts, or regulators. This within-subject design was utilized given the difficulty and expense of using professionals as experiment participants. As this element of the design suggests a possible confound in exploring treatment effects (see Davis and Holt 1994), this session is best viewed as a robustness check on results from student sessions. The session was conducted remotely from the participant's own computer through an Internet browser. Participants were divided into three separate markets of identical structure, each with six players. Experiment instructions were emailed just before the beginning of each treatment, and moderated through a conference call. The session lasted about 2 h, and average earnings were approximately $200.

All participants went through two practice sessions where they competed against five adaptive computer agents in both a uniform price and a discriminatory price auction. The user interface and many of the design parameters in the training sessions were exactly as they were in actual experiments. Participants were required to reach a rather modest earnings goal in a training session and they had to repeat the exercise until this goal was met. The purpose of the practice sessions was to allow participants

to develop and test various offer strategies as well as to minimize confusion and careless behavior in the paid auctions. In addition, the students were given more direct experience through uniform price and a discriminatory auction periods with other human participants (15 periods for each auction). Our experience with testing auctions suggests that running training sessions with other individuals, particularly for discriminatory auctions, is the most effective way for participants to become familiar with the software and auction pricing rules. Although the professionals did not undergo the 30 practice periods with other human participants, they did participate in four treatments prior to the two treatments reported here.[10]

4 Data and analysis

Figure 4 presents the average time series for the soft-cap auctions conducted with professionals. The soft-cap and uniform price auction results for students are presented in Figs. 5 and 6, respectively. As a point of reference, the figures also display marginal cost, defined as the true marginal cost of producing the last megawatt-hour needed to meet demand, which can be thought of as the competitive price.[11] Actual prices for all groups and all experiments are in general substantially above these competitive levels, particularly in low cost periods.

A few patterns emerge. First, for both students and professionals, prices tend to be constant in soft cap treatments. Although load is uncertain when offers are submitted, load varies across periods, and marginal costs vary, prices appear to show little sensitivity to this. There is more variation in prices in the uniform price auctions with students, and at least with price-responsive load there appears to be a positive relationship between price and marginal cost. To facilitate tests of Hypotheses 1 and 2 we turn to a formal analysis that adequately controls for experimental design factors.

Market price data from our experiment can be characterized as time-series cross-section (TSCS) data, which describes a panel where there are few cross-section units relative to time periods (T). For a particular treatment, each group is considered a cross-section unit observed over a 25-period (professional subjects) or 50-period (student subjects) time horizon. Market prices in a particular treatment are assumed to be a linear function of an intercept, marginal cost (*Marginal Cost*, in S/MWh), as defined above, and the load served for a particular period (*Load*, in MW). Formally, the regression model is

$$Price_{it} = \alpha_m + \beta_{1,m}\, Marginal\, Cost_{it} + \beta_{2,m}\, Load_{it} + \varepsilon_{it}$$

where $Price_{it}$ is the market price for group i in period t; m denotes treatment; ε_{it} is a mean-zero error term and the α_m, $\beta_{1,m}$, and $\beta_{2,m}$ are estimable parameters. A separate

[10] In particular, the professionals also participated in all four experimental auctions described above, with the exception that all periods involved the baseline (i.e., minimum) generation costs.

[11] For the uniform price auction treatments, and the soft-cap treatments in periods where costs were below the soft cap, this would have been the prevailing market price if everyone submitted offers equal to costs. In the soft-cap treatments, if some accepted offers are above the soft cap, the average market price would be lower than the "competitive" price.

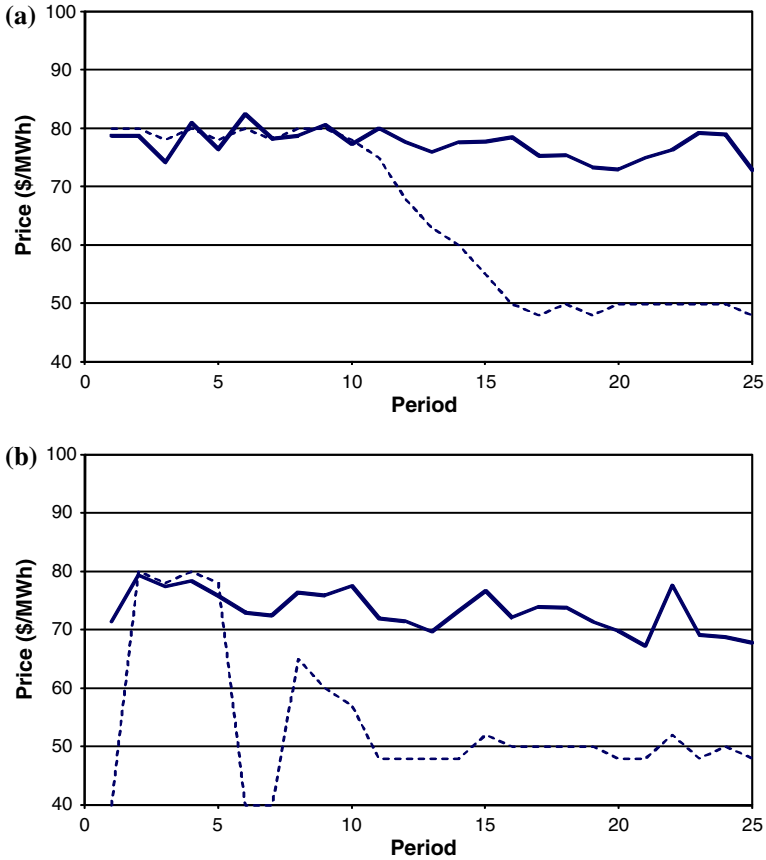

Fig. 4 Average market price per period (Professional Subjects) for soft cap auctions. **a** Without price-responsive load. **b** With price-responsive load. *Note*: Dotted line denotes the quasi-competitive price: the cost of generating the last MWh needed to meet demand when offers equal marginal cost

regression is estimated for students and professionals, but for each participant type data across all treatments are pooled. Treatment-specific interaction variables are utilized to allow for the intercept and slope coefficients to vary by treatment.

We allow regression errors to be correlated across time periods, due to factors such as learning, and assume they follow a first-order autoregressive (AR1) process that is common across groups.[12] To correct for autocorrelation we estimate the model parameters using the Prais-Winsten FGLS estimation procedure for panel data.[13] "Robust" panel-corrected standard errors are estimated (Beck and Katz 1995). Both the

[12] Beck and Katz (1996) advocate specifying a common AR(1) process versus group-specific AR(1) processes.

[13] This is done in two steps. The first step involves estimating a common autocorrelation coefficient using the estimated errors from an OLS regression. The second step uses the estimated autocorrelation coefficient to transform the dependent and independent variables in a second OLS regression to estimate the unknown model parameters.

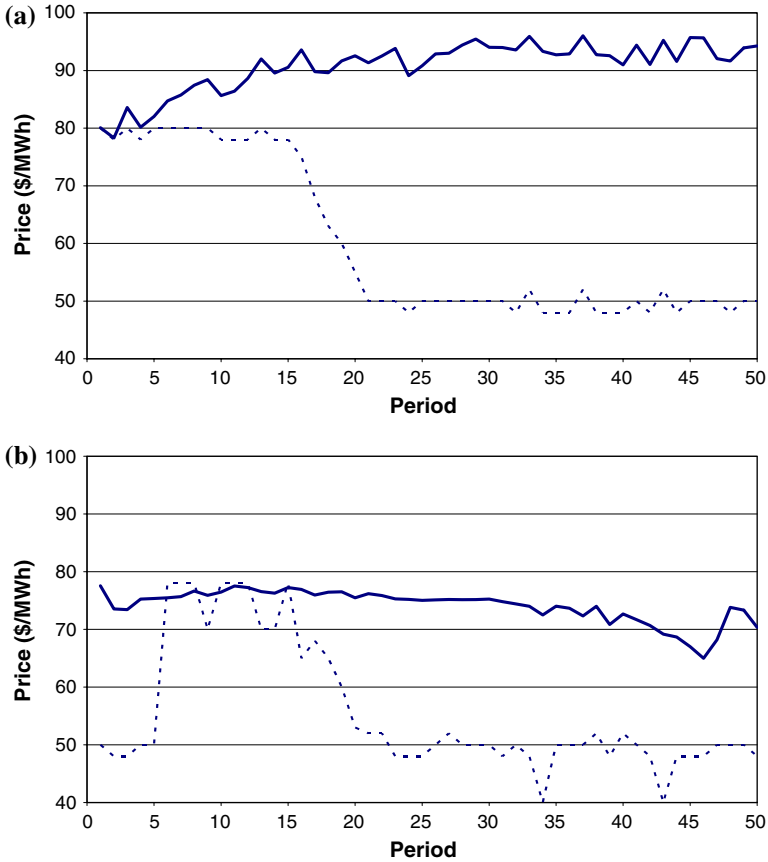

Fig. 5 Average market price per period (Student Subjects) for soft cap auctions. **a** Without price-responsive load. **b** With price-responsive load. *Note*: Dotted line denotes the quasi-competitive price: the cost of generating the last MWh needed to meet demand when offers equal marginal cost

coefficient estimator and the estimator of the asymptotic covariance matrix are consistent in T.

The estimated models are presented in Table 1. Focusing first on Model I (student subjects), *Load* is a statistically significant factor except for the case of the soft cap auction with price-responsive load. Consistent with expectations, the effect of load is more pronounced in the uniform price auction. The coefficients on *Load* are over twice as large in comparable uniform price auctions. This responsiveness to demand is consistent with Hypothesis 1. However, evidence in favor of Hypothesis 1 is indeed mixed. That is, although there is relatively small within-experiment variation with respect to load in soft cap auctions, the introduction of price-responsive load in and of itself had a marked impact. Estimated prices at approximately the mean of the design parameters (*Marginal Cost* = 60; *Load* = 490) are: \$90.59 (std. err. 0.98) for the soft cap; \$74.60 (1.02) for soft cap with price-responsive load; \$89.27 (1.00) for uniform price; and \$77.60 (1.01) for uniform price with price-responsive load.

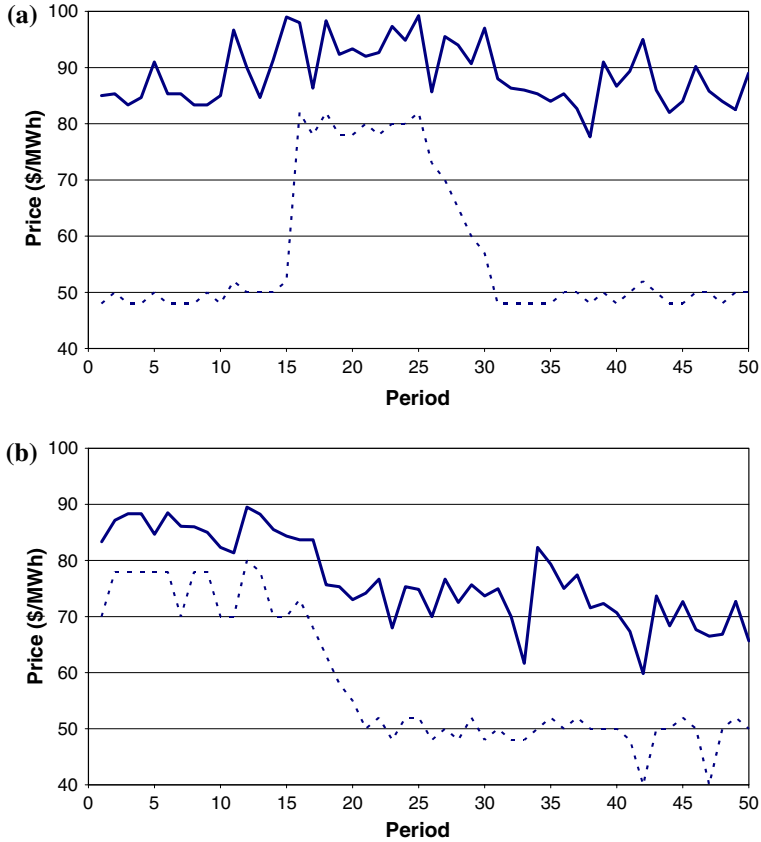

Fig. 6 Average market price per period (Student Subjects) for uniform price auctions. **a** Without price-responsive load. **b** With price-responsive load. *Note*: Dotted line denotes the quasi-competitive price: the cost of generating the last MWh needed to meet demand when offers equal marginal cost

Thus, for both auctions, prices drop about \$15 with price-responsive load, and these differences are statistically significant (soft cap: $z = 11.32$, $p < 0.01$; uniform price: $z = 8.22$, $p < 0.01$).

Turning to the effect of cost, the *Marginal Cost* coefficient for the soft cap auction has the wrong sign and is statistically significant, whereas for the soft cap auction with price-responsive load the coefficient is not statistically significant.[14] In contrast the effect of cost in the two uniform price treatments is statistically significant and has the expected sign. Relative to the comparable soft cap treatment, the cost effect

[14] As a crude control for learning, we also estimated the model with treatment-specific time trends. For the soft cap auction, the coefficient on *Marginal Cost* remains negative but is no longer statistically significant, and the time trend is positive and statistically significant. The students may have still been learning how to exploit their market power at the beginning of the experiment, and given that marginal cost falls towards the end of the experiment, when the time trend is excluded the *Marginal Cost* variable may act as an (inverse) proxy for learning. We note that the conclusions regarding our two main hypotheses are not affected by inclusion or exclusion of time trends.

Table 1 Analysis of prices from experimental auction markets for electric power

Variable	Model I: Students	Model II: Professionals
Soft-cap auction		
Intercept	82.550*	47.311*
	(4.723)	(4.812)
Marginal cost	−0.235*	0.089
	(0.067)	(0.058)
Load	0.045*	0.049*
	(0.007)	(0.008)
Soft cap auction with price-responsive load		
Intercept	63.489*	50.659*
	(4.588)	(4.032)
Marginal cost	0.096	0.104*
	(0.061)	(0.038)
Load	0.011	0.037*
	(0.008)	(0.010)
Uniform price auction		
Intercept	38.693*	
	(4.065)	
Marginal cost	0.160*	
	(0.058)	
Load	0.084*	
	(0.007)	
Uniform price auction with price-responsive load		
Intercept	33.551*	
	(4.305)	
Marginal cost	0.499*	
	(0.062)	
Load	0.029*	
	(0.008)	
ρ	0.683	0.623
# Observations	600	150
R^2	0.980	0.991

Notes: Panel-corrected standard errors in parentheses. * Denotes coefficient is statistically different than zero at the 5% significance level

is statistically different and larger for the uniform price auction with either inelastic demand ($z = 4.44$, $p < 0.01$) or price-responsive load ($z = 4.63$, $p < 0.01$). This evidence supports our Hypothesis 2. For the uniform price auction, we further note that the cost effect is more pronounced with price-responsive load; indeed the coefficient with price-responsive load is more than three times as large and is statistically different than with inelastic load ($z = 3.99$, $p < 0.01$).

In the two soft cap auction treatments with professionals (Model II), all estimated parameters are statistically significant except for the coefficient on *Marginal Cost* for the soft cap auction with inelastic load. The overall results provide support of Hypothesis 1. Estimated prices at approximately the mean of the design parameters (*Marginal Cost*=60; *Load*=490) are \$74.97 (0.91) and \$76.63 (0.88), respectively, with and without price-responsive load. These figures are not statistically different ($z = 1.31$, $p = 0.19$). Within-treatment variation with respect to changes in marginal cost is rather low. In particular, for either treatment, price increases by about \$0.10 for every \$1 increase in marginal cost. To put this into perspective, this amounts to about a \$4 difference between the lowest (\$40) and highest (\$80) levels of marginal cost encountered. As is evident from comparing Figs. 4 and 5, students were much more successful at realizing high prices well above the soft cap when load is stochastic but

inelastic ($90.59 vs. $76.63). Prices with price-responsive load are quite close for the two participant groups ($74.60 vs. $74.97).

Aside from the nature of the participants, the soft cap treatments with professionals differed in that they were part of a within-subject design and treatments were run in a single session. The initial prices in the two soft cap experiments with students were both around $80, but with inelastic load students were able to drive up prices beyond $90 in most periods. This trend in prices occurred in all three sessions. An interesting yet important question is whether the high prices would have persisted had we had students in the soft cap auction with inelastic demand, similar to the professionals, transition to a price-responsive load setting. Individual offer data suggests that in the price-responsive load setting, student participants simply did not make the same attempt to increase prices. Instead, simply knowing about the price-responsive load program appears to have had an effect. This coupled with our findings that soft cap auction prices are fairly insensitive to generation costs and fluctuations in load does suggest that modifying our experimental design with students would have produced different results.

5 Conclusion

This paper adds to the small literature that studies the design of wholesale markets for electricity through a controlled auction experiment with human participants. In particular, we analyze the effect of including a "soft" price cap in a uniform price auction, in situations with and without price-responsive load. This soft cap converts the standard uniform price auction to a hybrid auction mechanism whereby offers below the soft cap are submitted into a uniform price auction and offers above the cap are submitted into a discriminatory or pay-as-offered auction. A soft cap was implemented in December 2000 in California, in the midst of the electricity crisis, and remains in effect for the Western Electricity Coordinating Council.

Although our experiment abstracts from some characteristics of the California wholesale markets during the electricity crisis, for instance there was no opportunity for arbitrage through participation in day-ahead and spot markets, our results suggest that California may have been better off not implementing the soft cap. As in previous experiments with pure discriminatory auctions, the offer curves in the soft-cap auctions tended to be relatively flat. As an example, in the last period from the soft cap auction with inelastic load, the difference between the lowest and highest submitted offer was less than $10 for all three groups of students. The corresponding difference could be $80 in a uniform price auction (the price cap of $100/MWh minus the lowest cost of generation). The flat offer curves make the market price relatively insensitive to reductions of load compared to the hockey stick shape in a uniform price auction. This may explain why load reductions were relatively ineffective in California when the soft-cap auction was in effect. Prices remained at high levels from January to May 2001 in spite of major efforts to conserve power.

The California spot market currently uses a soft cap auction in tandem with an AMP. Absent a strong interaction effect our findings along with Kiesling and Wilson (2007), who investigate experimentally a California-like AMP with a uniform price

auction without a soft price cap, paint an interesting picture. Kiesling and Wilson (2007) find that the AMP has little effect on prices regardless of the level of demand or market power. As we find soft cap auction prices are relatively insensitive to load or generation costs, the two studies together cast doubt that California's price mitigation measures have a marked affect on prices. Invariance to such fundamental market characteristics is indeed indicative of a flawed market mechanism. The evidence from the body of experimental work supports the conclusion that the development of a two-sided, uniform price auction without a soft cap is the best way to reach the goal of competitive wholesale markets for electricity.

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